Lab 2: Hidden Markov Models

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Question 1: Build the HMM

Implementation: Built 10x10 transition and emission matrices: each sector stays or moves one step forward (wrapping 10 to 1) with probability 0.5, and the sensor reports one of sectors $i-2, \ldots, i+2$ (modulo 10) with probability 0.2 each.

```
# --- Build HMM matrices: robot on a ring of 10 sectors ---
# Hidden states = true sector, symbols = sensor reading (both 1..10)
states <- 1:10
symbols <- 1:10

# transProbs[from, to]: rows = current state z_t, cols = next state z_t+1
#   (each row sums to 1)
# emissionProbs[state, symbol]: rows = hidden state, cols = observed symbol
#   (each row sums to 1)
transProbs <- matrix(0,10,10)
emissionProbs <- matrix(0,10,10)

for(i in 1:10) {
  # Stay in sector i or move to i+1, each w.p. 0.5.
  # (i %% 10) + 1 wraps 10 -> 1 (ring)
  transProbs[i,i] <- 0.5
  transProbs[i, (i%%10) + 1] <- 0.5
  # Sensor reports one of i-2..i+2 uniformly (5 values -> 0.2 each).
  # ((x - 1) %% 10) + 1 maps any integer onto 1..10, e.g. 0 -> 10, -1 -> 9
  for(k in -2:2){
    emissionProbs[i, ((i+k-1) %% 10) + 1] <- 0.2
  }
}
```

Question 2: Simulate 100 time steps

Implementation: Created the model with initHMM() using a uniform start distribution (0.1 per sector), then drew 100 hidden states and observations with simHMM() (no seed set).

```
# --- Create HMM object and simulate ---
# initHMM(States, Symbols, startProbs, transProbs, emissionProbs);
# startProbs = uniform over the 10 sectors
hmm <- initHMM(States = states, Symbols = symbols, transProbs = transProbs,
               emissionProbs = emissionProbs, startProbs = rep(0.1, 10))
```

```r
# simHMM returns a list: $states (true hidden path), $observation (readings)
simulation <- simHMM(hmm, 100)
```

Question 3: Filtered, smoothed, and most probable path

Implementation: Ran forward() and backward() from the HMM package; per time step, exponentiated the log-probabilities (after subtracting the column max) and normalised. Filtered uses alpha, smoothed uses alpha times beta, and the most probable path comes from viterbi().

```r
# --- Filtering, smoothing (forward-backward) and Viterbi ---
observations <- simulation$observation

filtered <- rep(0, 100)
smoothed <- rep(0, 100)

# forward/backward return LOG probabilities, matrix dims = states x time
#   logAlpha[i, t] = log p(x_0:t, z_t = i)
#   logBeta[i, t]  = log p(x_t+1:T | z_t = i)
logAlpha <- forward(hmm, observations)
logBeta <- backward(hmm, observations)

# filtered: p(z_t | x_0:t) propto alpha_t. Per column (time step):
# subtract max before exp to avoid underflow, then normalize to sum 1.
# (Log scale because alpha is a product of many probabilities < 1 and
# underflows to 0 for long sequences. exp(col - max) = alpha / max(alpha):
# the largest entry becomes 1 and the constant cancels when normalizing.)
# Result: 10 x 100 matrix, column t = distribution over states at t
filtered <- apply(logAlpha, 2, function(col){
  col <- exp(col - max(col))
  col/sum(col)
})

# smoothed: p(z_t | x_0:T) propto alpha_t * beta_t (sum in log space)
smoothed <- apply(logAlpha + logBeta, 2, function(col){
  col <- exp(col - max(col))
  col/sum(col)
})

# probable path: Viterbi = most probable JOINT state sequence
# argmax_{z_0:T} p(z_0:T | x_0:T); returns vector of states (length T)
probablePath <- viterbi(hmm, observations)

#prints
print(filtered)
```

```
##       index
## states   1         2         3         4         5         6          7
##      1  0.2 0.0000000 0.3333333 0.5000000 0.7142857 0.5000000 0.32142857
##      2  0.0 0.0000000 0.0000000 0.1666667 0.0000000 0.3571429 0.42857143
##      3  0.0 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.17857143
##      4  0.0 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.00000000
##      5  0.0 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.00000000
```

```
##     6  0.0 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.00000000
##     7  0.2 0.1428571 0.0000000 0.0000000 0.0000000 0.0000000 0.00000000
##     8  0.2 0.2857143 0.0000000 0.0000000 0.0000000 0.0000000 0.00000000
##     9  0.2 0.2857143 0.0000000 0.0000000 0.0000000 0.0000000 0.00000000
##     10 0.2 0.2857143 0.6666667 0.3333333 0.2857143 0.1428571 0.07142857
##       index
## states          8         9        10        11         12        13        14
##     1  0.32352941 0.1969697 0.0000000 0.0000000 0.00000000 0.0000000 0.0000000
##     2  0.61764706 0.4848485 0.3781513 0.0000000 0.00000000 0.0000000 0.0000000
##     3  0.00000000 0.3181818 0.4453782 0.5077720 0.00000000 0.0000000 0.0000000
##     4  0.00000000 0.0000000 0.1764706 0.3834197 0.59722222 0.3099099 0.1549550
##     5  0.00000000 0.0000000 0.0000000 0.1088083 0.32986111 0.4810811 0.3954955
##     6  0.00000000 0.0000000 0.0000000 0.0000000 0.07291667 0.2090090 0.3450450
##     7  0.00000000 0.0000000 0.0000000 0.0000000 0.00000000 0.0000000 0.1045045
##     8  0.00000000 0.0000000 0.0000000 0.0000000 0.00000000 0.0000000 0.0000000
##     9  0.00000000 0.0000000 0.0000000 0.0000000 0.00000000 0.0000000 0.0000000
##     10 0.05882353 0.0000000 0.0000000 0.0000000 0.00000000 0.0000000 0.0000000
##       index
## states        15         16         17        18        19        20        21
##     1  0.0000000 0.00000000 0.00000000 0.0000000 0.0000000 0.0000000 0.2738399
##     2  0.0000000 0.00000000 0.00000000 0.0000000 0.0000000 0.0000000 0.0000000
##     3  0.0000000 0.00000000 0.00000000 0.0000000 0.0000000 0.0000000 0.0000000
##     4  0.1071651 0.05358255 0.02679128 0.0000000 0.0000000 0.0000000 0.0000000
##     5  0.3806854 0.24392523 0.14875389 0.0000000 0.0000000 0.0000000 0.0000000
##     6  0.5121495 0.44641745 0.34517134 0.0000000 0.0000000 0.0000000 0.0000000
##     7  0.0000000 0.25607477 0.35124611 0.0000000 0.0000000 0.0000000 0.0000000
##     8  0.0000000 0.00000000 0.12803738 0.7891767 0.0000000 0.0000000 0.0000000
##     9  0.0000000 0.00000000 0.00000000 0.2108233 0.8258843 0.4523202 0.2261601
##     10 0.0000000 0.00000000 0.00000000 0.0000000 0.1741157 0.5476798 0.5000000
##       index
## states 22  23   24    25  26  27     28        29        30        31        32
##     1   0 0.0 0.00 0.000 0.0 0.0 0.0000 0.0000000 0.0000000 0.0000000 0.0000000
##     2   1 0.5 0.25 0.125 0.0 0.0 0.0000 0.0000000 0.0000000 0.0000000 0.0000000
##     3   0 0.5 0.50 0.375 0.0 0.0 0.0000 0.0000000 0.0000000 0.0000000 0.0000000
##     4   0 0.0 0.25 0.375 0.0 0.0 0.0000 0.0000000 0.0000000 0.0000000 0.0000000
##     5   0 0.0 0.00 0.125 0.8 0.4 0.0000 0.0000000 0.0000000 0.0000000 0.0000000
##     6   0 0.0 0.00 0.000 0.2 0.5 0.5625 0.2903226 0.1636364 0.0000000 0.0000000
##     7   0 0.0 0.00 0.000 0.0 0.1 0.3750 0.4838710 0.4363636 0.3267327 0.0000000
##     8   0 0.0 0.00 0.000 0.0 0.0 0.0625 0.2258065 0.4000000 0.4554455 0.0000000
##     9   0 0.0 0.00 0.000 0.0 0.0 0.0000 0.0000000 0.0000000 0.2178218 0.7555556
##     10  0 0.0 0.00 0.000 0.0 0.0 0.0000 0.0000000 0.0000000 0.0000000 0.2444444
##       index
## states 33  34   35    36     37     38  39  40        41        42 43 44 45  46
##     1   0 0.0 0.25 0.375 0.3750 0.0000 0.0 0.0 0.0000000 0.0000000  0  0  0 0.0
##     2   0 0.0 0.00 0.125 0.2500 0.6250 0.4 0.2 0.0000000 0.0000000  0  0  0 0.0
##     3   0 0.0 0.00 0.000 0.0625 0.3125 0.6 0.5 0.3888889 0.2121212  0  0  0 0.0
##     4   0 0.0 0.00 0.000 0.0000 0.0625 0.0 0.3 0.4444444 0.4545455  0  0  0 0.0
##     5   0 0.0 0.00 0.000 0.0000 0.0000 0.0 0.0 0.1666667 0.3333333  0  0  0 0.0
##     6   0 0.0 0.00 0.000 0.0000 0.0000 0.0 0.0 0.0000000 0.0000000  1  1  0 0.0
##     7   0 0.0 0.00 0.000 0.0000 0.0000 0.0 0.0 0.0000000 0.0000000  0  0  1 0.5
##     8   0 0.0 0.00 0.000 0.0000 0.0000 0.0 0.0 0.0000000 0.0000000  0  0  0 0.5
##     9   1 0.5 0.25 0.125 0.0625 0.0000 0.0 0.0 0.0000000 0.0000000  0  0  0 0.0
##     10  0 0.5 0.50 0.375 0.2500 0.0000 0.0 0.0 0.0000000 0.0000000  0  0  0 0.0
##       index
```

```
## states        47        48  49 50 51  52 53  54 55 56 57  58        59   60
##     1  0.0000000 0.0000000 0.0  0  0 0.5  0 0.0  0  0  0 0.0 0.0000000 0.00
##     2  0.0000000 0.0000000 0.0  0  0 0.0  1 0.5  1  1  1 0.5 0.0000000 0.00
##     3  0.0000000 0.0000000 0.0  0  0 0.0  0 0.5  0  0  0 0.5 0.6666667 0.00
##     4  0.0000000 0.0000000 0.0  0  0 0.0  0 0.0  0  0  0 0.0 0.3333333 0.75
##     5  0.0000000 0.0000000 0.0  0  0 0.0  0 0.0  0  0  0 0.0 0.0000000 0.25
##     6  0.0000000 0.0000000 0.0  0  0 0.0  0 0.0  0  0  0 0.0 0.0000000 0.00
##     7  0.3333333 0.1666667 0.2  0  0 0.0  0 0.0  0  0  0 0.0 0.0000000 0.00
##     8  0.6666667 0.5000000 0.8  0  0 0.0  0 0.0  0  0  0 0.0 0.0000000 0.00
##     9  0.0000000 0.3333333 0.0  1  0 0.0  0 0.0  0  0  0 0.0 0.0000000 0.00
##     10 0.0000000 0.0000000 0.0  0  1 0.5  0 0.0  0  0  0 0.0 0.0000000 0.00
##       index
## states    61         62        63        64        65       66        67
##     1  0.000 0.00000000 0.0000000 0.0000000 0.0000000 0.000000 0.0000000
##     2  0.000 0.00000000 0.0000000 0.0000000 0.0000000 0.000000 0.0000000
##     3  0.000 0.00000000 0.0000000 0.0000000 0.0000000 0.000000 0.0000000
##     4  0.375 0.00000000 0.0000000 0.0000000 0.0000000 0.000000 0.0000000
##     5  0.500 0.53846154 0.3684211 0.2692308 0.1346154 0.000000 0.0000000
##     6  0.125 0.38461538 0.6315789 0.7307692 0.5000000 0.000000 0.0000000
##     7  0.000 0.07692308 0.0000000 0.0000000 0.3653846 0.703125 0.3515625
##     8  0.000 0.00000000 0.0000000 0.0000000 0.0000000 0.296875 0.5000000
##     9  0.000 0.00000000 0.0000000 0.0000000 0.0000000 0.000000 0.1484375
##     10 0.000 0.00000000 0.0000000 0.0000000 0.0000000 0.000000 0.0000000
##       index
## states        68         69         70        71       72       73        74
##     1  0.0000000 0.00000000 0.00000000 0.1632047 0.765158 0.382579 0.1912895
##     2  0.0000000 0.00000000 0.00000000 0.0000000 0.234842 0.500000 0.4412895
##     3  0.0000000 0.00000000 0.00000000 0.0000000 0.000000 0.117421 0.3087105
##     4  0.0000000 0.00000000 0.00000000 0.0000000 0.000000 0.000000 0.0587105
##     5  0.0000000 0.00000000 0.00000000 0.0000000 0.000000 0.000000 0.0000000
##     6  0.0000000 0.00000000 0.00000000 0.0000000 0.000000 0.000000 0.0000000
##     7  0.1898734 0.09493671 0.05202312 0.0000000 0.000000 0.000000 0.0000000
##     8  0.4599156 0.32489451 0.23005780 0.1448071 0.000000 0.000000 0.0000000
##     9  0.3502110 0.40506329 0.40000000 0.3234421 0.000000 0.000000 0.0000000
##     10 0.0000000 0.17510549 0.31791908 0.3685460 0.000000 0.000000 0.0000000
##       index
## states        75         76        77        78        79        80        81
##     1  0.1215410 0.06077048 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000
##     2  0.4019262 0.26173359 0.1663053 0.1183749 0.0000000 0.0000000 0.0000000
##     3  0.4765328 0.43922952 0.3614647 0.3756628 0.2625590 0.0000000 0.0000000
##     4  0.0000000 0.23826641 0.3493635 0.5059623 0.4685445 0.0000000 0.0000000
##     5  0.0000000 0.00000000 0.1228665 0.0000000 0.2688964 0.7327969 0.0000000
##     6  0.0000000 0.00000000 0.0000000 0.0000000 0.0000000 0.2672031 0.7891395
##     7  0.0000000 0.00000000 0.0000000 0.0000000 0.0000000 0.0000000 0.2108605
##     8  0.0000000 0.00000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000
##     9  0.0000000 0.00000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000
##     10 0.0000000 0.00000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000
##       index
## states        82       83        84         85        86        87         88
##     1  0.0000000 0.000000 0.0000000 0.00000000 0.0000000 0.0000000 0.09515177
##     2  0.0000000 0.000000 0.0000000 0.00000000 0.0000000 0.0000000 0.00000000
##     3  0.0000000 0.000000 0.0000000 0.00000000 0.0000000 0.0000000 0.00000000
##     4  0.0000000 0.000000 0.0000000 0.00000000 0.0000000 0.0000000 0.00000000
##     5  0.0000000 0.000000 0.0000000 0.00000000 0.0000000 0.0000000 0.00000000
```

```
##     6  0.4410721 0.220536 0.1281787 0.08285819 0.0000000 0.0000000 0.00000000
##     7  0.5589279 0.500000 0.4187858 0.35357274 0.0000000 0.0000000 0.00000000
##     8  0.0000000 0.279464 0.4530355 0.56356906 0.6193929 0.3096965 0.15484823
##     9  0.0000000 0.000000 0.0000000 0.00000000 0.3806071 0.5000000 0.40484823
##     10 0.0000000 0.000000 0.0000000 0.00000000 0.0000000 0.1903035 0.34515177
##       index
## states         89        90 91  92        93        94  95        96        97
##     1  0.23862728 0.0000000  0 0.0 0.0000000 0.0000000 0.0 0.0000000 0.0000000
##     2  0.05156853 0.8491108  1 0.5 0.3333333 0.1666667 0.1 0.0000000 0.0000000
##     3  0.00000000 0.1508892  0 0.5 0.6666667 0.5000000 0.4 0.2631579 0.0000000
##     4  0.00000000 0.0000000  0 0.0 0.0000000 0.3333333 0.5 0.4736842 0.4242424
##     5  0.00000000 0.0000000  0 0.0 0.0000000 0.0000000 0.0 0.2631579 0.4242424
##     6  0.00000000 0.0000000  0 0.0 0.0000000 0.0000000 0.0 0.0000000 0.1515152
##     7  0.00000000 0.0000000  0 0.0 0.0000000 0.0000000 0.0 0.0000000 0.0000000
##     8  0.00000000 0.0000000  0 0.0 0.0000000 0.0000000 0.0 0.0000000 0.0000000
##     9  0.30333356 0.0000000  0 0.0 0.0000000 0.0000000 0.0 0.0000000 0.0000000
##     10 0.40647063 0.0000000  0 0.0 0.0000000 0.0000000 0.0 0.0000000 0.0000000
##       index
## states         98 99 100
##     1  0.00000000  0 0.0
##     2  0.00000000  0 0.0
##     3  0.00000000  0 0.0
##     4  0.00000000  0 0.0
##     5  0.53846154  0 0.0
##     6  0.36538462  0 0.0
##     7  0.09615385  0 0.0
##     8  0.00000000  1 0.5
##     9  0.00000000  0 0.5
##     10 0.00000000  0 0.0
```

```
print(smoothed)
```

```
##       index
## states         1         2         3         4         5         6          7
##     1  0.0000000 0.0000000 0.1311163 0.3933489 0.6555814 0.6553696 0.50518958
##     2  0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.1874603 0.44990468
##     3  0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.00000000
##     4  0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.00000000
##     5  0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.00000000
##     6  0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.00000000
##     7  0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.00000000
##     8  0.2172209 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.00000000
##     9  0.5000000 0.4344419 0.0000000 0.0000000 0.0000000 0.0000000 0.00000000
##     10 0.2827791 0.5655581 0.8688837 0.6066511 0.3444186 0.1571701 0.04490574
##       index
## states          8          9        10         11         12        13
##     1  0.205041305 0.04956577 0.0000000 0.00000000 0.00000000 0.0000000
##     2  0.787333192 0.47447575 0.1715738 0.00000000 0.00000000 0.0000000
##     3  0.000000000 0.47595848 0.5837746 0.37364965 0.00000000 0.0000000
##     4  0.000000000 0.00000000 0.2446516 0.53293794 0.65579326 0.2550307
##     5  0.000000000 0.00000000 0.0000000 0.09341241 0.32196569 0.6221140
##     6  0.000000000 0.00000000 0.0000000 0.00000000 0.02224105 0.1228553
##     7  0.000000000 0.00000000 0.0000000 0.00000000 0.00000000 0.0000000
##     8  0.000000000 0.00000000 0.0000000 0.00000000 0.00000000 0.0000000
```

```
##     9  0.000000000 0.00000000 0.0000000 0.00000000 0.00000000 0.0000000
##     10 0.007625503 0.00000000 0.0000000 0.00000000 0.00000000 0.0000000
##       index
## states         14        15        16       17        18         19 20 21 22
##     1  0.00000000 0.0000000 0.0000000 0.000000 0.0000000 0.0000000  0  1  0
##     2  0.00000000 0.0000000 0.0000000 0.000000 0.0000000 0.0000000  0  0  1
##     3  0.00000000 0.0000000 0.0000000 0.000000 0.0000000 0.0000000  0  0  0
##     4  0.03643296 0.0000000 0.0000000 0.000000 0.0000000 0.0000000  0  0  0
##     5  0.55793264 0.1294217 0.0000000 0.000000 0.0000000 0.0000000  0  0  0
##     6  0.40563440 0.8705783 0.3035374 0.000000 0.0000000 0.0000000  0  0  0
##     7  0.00000000 0.0000000 0.6964626 0.477653 0.0000000 0.0000000  0  0  0
##     8  0.00000000 0.0000000 0.0000000 0.522347 0.6517687 0.0000000  0  0  0
##     9  0.00000000 0.0000000 0.0000000 0.000000 0.3482313 0.8258843  0  0  0
##     10 0.00000000 0.0000000 0.0000000 0.000000 0.0000000 0.1741157  1  0  0
##       index
## states        23        24        25        26        27          28         29
##     1  0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.00000000 0.0000000
##     2  0.1911765 0.0000000 0.0000000 0.0000000 0.0000000 0.00000000 0.0000000
##     3  0.8088235 0.3823529 0.0000000 0.0000000 0.0000000 0.00000000 0.0000000
##     4  0.0000000 0.6176471 0.5735294 0.0000000 0.0000000 0.00000000 0.0000000
##     5  0.0000000 0.0000000 0.4264706 0.7647059 0.2352941 0.00000000 0.0000000
##     6  0.0000000 0.0000000 0.0000000 0.2352941 0.6617647 0.52941176 0.1323529
##     7  0.0000000 0.0000000 0.0000000 0.0000000 0.1029412 0.44117647 0.6617647
##     8  0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.02941176 0.2058824
##     9  0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.00000000 0.0000000
##     10 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.00000000 0.0000000
##       index
## states        30        31 32 33        34         35        36         37
##     1  0.0000000 0.0000000  0  0 0.0000000 0.38181818 0.6000000 0.43636364
##     2  0.0000000 0.0000000  0  0 0.0000000 0.00000000 0.1818182 0.50909091
##     3  0.0000000 0.0000000  0  0 0.0000000 0.00000000 0.0000000 0.05454545
##     4  0.0000000 0.0000000  0  0 0.0000000 0.00000000 0.0000000 0.00000000
##     5  0.0000000 0.0000000  0  0 0.0000000 0.00000000 0.0000000 0.00000000
##     6  0.0000000 0.0000000  0  0 0.0000000 0.00000000 0.0000000 0.00000000
##     7  0.3529412 0.0000000  0  0 0.0000000 0.00000000 0.0000000 0.00000000
##     8  0.6470588 0.6764706  0  0 0.0000000 0.00000000 0.0000000 0.00000000
##     9  0.0000000 0.3235294  1  1 0.3454545 0.07272727 0.0000000 0.00000000
##     10 0.0000000 0.0000000  0  0 0.6545455 0.54545455 0.2181818 0.00000000
##       index
## states        38        39        40        41 42 43 44 45   46  47   48 49 50
##     1  0.0000000 0.0000000 0.0000000 0.0000000  0  0  0  0 0.00 0.0 0.00  0  0
##     2  0.7272727 0.1818182 0.0000000 0.0000000  0  0  0  0 0.00 0.0 0.00  0  0
##     3  0.2727273 0.8181818 0.4545455 0.0000000  0  0  0  0 0.00 0.0 0.00  0  0
##     4  0.0000000 0.0000000 0.5454545 0.7272727  0  0  0  0 0.00 0.0 0.00  0  0
##     5  0.0000000 0.0000000 0.0000000 0.2727273  1  0  0  0 0.00 0.0 0.00  0  0
##     6  0.0000000 0.0000000 0.0000000 0.0000000  0  1  1  0 0.00 0.0 0.00  0  0
##     7  0.0000000 0.0000000 0.0000000 0.0000000  0  0  0  1 0.75 0.5 0.25  0  0
##     8  0.0000000 0.0000000 0.0000000 0.0000000  0  0  0  0 0.25 0.5 0.75  1  0
##     9  0.0000000 0.0000000 0.0000000 0.0000000  0  0  0  0 0.00 0.0 0.00  0  1
##     10 0.0000000 0.0000000 0.0000000 0.0000000  0  0  0  0 0.00 0.0 0.00  0  0
##       index
## states 51 52 53 54 55 56 57        58        59        60         61        62
##     1   0  1  0  0  0  0  0 0.0000000 0.0000000 0.0000000 0.00000000 0.0000000
##     2   0  0  1  1  1  1  1 0.2655989 0.0000000 0.0000000 0.00000000 0.0000000
```

```
##     3   0  0  0  0  0  0  0 0.7344011 0.5311978 0.0000000 0.00000000 0.0000000
##     4   0  0  0  0  0  0  0 0.0000000 0.4688022 0.7967967 0.32799458 0.0000000
##     5   0  0  0  0  0  0  0 0.0000000 0.0000000 0.2032033 0.62506956 0.7653207
##     6   0  0  0  0  0  0  0 0.0000000 0.0000000 0.0000000 0.04693586 0.2346793
##     7   0  0  0  0  0  0  0 0.0000000 0.0000000 0.0000000 0.00000000 0.0000000
##     8   0  0  0  0  0  0  0 0.0000000 0.0000000 0.0000000 0.00000000 0.0000000
##     9   0  0  0  0  0  0  0 0.0000000 0.0000000 0.0000000 0.00000000 0.0000000
##     10  1  0  0  0  0  0  0 0.0000000 0.0000000 0.0000000 0.00000000 0.0000000
##       index
## states        63        64        65        66        67         68        69
##     1  0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.00000000 0.0000000
##     2  0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.00000000 0.0000000
##     3  0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.00000000 0.0000000
##     4  0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.00000000 0.0000000
##     5  0.4367697 0.1082186 0.0000000 0.0000000 0.0000000 0.00000000 0.0000000
##     6  0.5632303 0.8917814 0.4019549 0.0000000 0.0000000 0.00000000 0.0000000
##     7  0.0000000 0.0000000 0.5980451 0.6956911 0.2377295 0.04253781 0.0000000
##     8  0.0000000 0.0000000 0.0000000 0.3043089 0.6513232 0.47279764 0.1455739
##     9  0.0000000 0.0000000 0.0000000 0.0000000 0.1109473 0.48466454 0.6513232
##     10 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.00000000 0.2031030
##       index
## states        70        71        72         73        74         75         76
##     1  0.0000000 0.4129782 0.8469752 0.46780605 0.2080506 0.06770877 0.01046408
##     2  0.0000000 0.0000000 0.1530248 0.49554356 0.5992350 0.46409910 0.24654827
##     3  0.0000000 0.0000000 0.0000000 0.03665039 0.1927144 0.46819213 0.60059932
##     4  0.0000000 0.0000000 0.0000000 0.00000000 0.0000000 0.00000000 0.14238833
##     5  0.0000000 0.0000000 0.0000000 0.00000000 0.0000000 0.00000000 0.00000000
##     6  0.0000000 0.0000000 0.0000000 0.00000000 0.0000000 0.00000000 0.00000000
##     7  0.0000000 0.0000000 0.0000000 0.00000000 0.0000000 0.00000000 0.00000000
##     8  0.0000000 0.0000000 0.0000000 0.00000000 0.0000000 0.00000000 0.00000000
##     9  0.3270685 0.0000000 0.0000000 0.00000000 0.0000000 0.00000000 0.00000000
##     10 0.6729315 0.5870218 0.0000000 0.00000000 0.0000000 0.00000000 0.00000000
##       index
## states         77       78        79        80        81        82        83
##     1  0.00000000 0.000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000
##     2  0.05553205 0.000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000
##     3  0.53959548 0.176231 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000
##     4  0.40487248 0.823769 0.4135882 0.0000000 0.0000000 0.0000000 0.0000000
##     5  0.00000000 0.000000 0.5864118 0.6509454 0.0000000 0.0000000 0.0000000
##     6  0.00000000 0.000000 0.0000000 0.3490546 0.8883025 0.4702778 0.1828858
##     7  0.00000000 0.000000 0.0000000 0.0000000 0.1116975 0.5297222 0.6515760
##     8  0.00000000 0.000000 0.0000000 0.0000000 0.0000000 0.0000000 0.1655382
##     9  0.00000000 0.000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000
##     10 0.00000000 0.000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000
##       index
## states         84        85        86        87        88        89 90 91  92
##     1  0.00000000 0.0000000 0.0000000 0.0000000 0.3554051 0.8222975  0  0 0.0
##     2  0.00000000 0.0000000 0.0000000 0.0000000 0.0000000 0.1777025  1  1 0.6
##     3  0.00000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000  0  0 0.4
##     4  0.00000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000  0  0 0.0
##     5  0.00000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000  0  0 0.0
##     6  0.02612654 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000  0  0 0.0
##     7  0.51216436 0.1114873 0.0000000 0.0000000 0.0000000 0.0000000  0  0 0.0
##     8  0.46170910 0.8885127 0.2891898 0.0000000 0.0000000 0.0000000  0  0 0.0
```

```
##      9  0.00000000 0.0000000 0.7108102 0.4668924 0.0000000 0.0000000  0  0 0.0
##      10 0.00000000 0.0000000 0.0000000 0.5331076 0.6445949 0.0000000  0  0 0.0
##        index
## states  93  94 95 96 97 98 99 100
##      1  0.0 0.0  0  0  0  0  0 0.0
##      2  0.2 0.0  0  0  0  0  0 0.0
##      3  0.8 0.6  0  0  0  0  0 0.0
##      4  0.0 0.4  1  0  0  0  0 0.0
##      5  0.0 0.0  0  1  0  0  0 0.0
##      6  0.0 0.0  0  0  1  0  0 0.0
##      7  0.0 0.0  0  0  0  1  0 0.0
##      8  0.0 0.0  0  0  0  0  1 0.5
##      9  0.0 0.0  0  0  0  0  0 0.5
##      10 0.0 0.0  0  0  0  0  0 0.0
```

```r
print(probablePath)
```

```
##   [1]  9 10  1  1  1  1  1  1  1  2  3  4  4  4  5  6  7  8  9 10  1  2  2  3  4
##  [26]  5  5  6  6  7  8  9  9 10  1  1  1  2  2  3  4  5  6  6  7  7  7  7  8  9
##  [51] 10  1  2  2  2  2  2  2  3  4  4  5  5  5  6  7  7  8  9 10  1  1  1  1  1
##  [76]  1  2  3  4  5  6  6  6  7  8  9 10  1  1  2  2  2  2  3  4  5  6  7  8  8
```

Theory: Three different inference targets, all using only the observations $x_{0:T}$:

- *Filtering* $p(z_t \mid x_{0:t})$: belief about the current state given readings so far (online). Forward recursion: $\alpha_t(z_t) = p(x_t \mid z_t) \sum_{z_{t-1}} p(z_t \mid z_{t-1})\, \alpha_{t-1}(z_{t-1})$, and $p(z_t \mid x_{0:t}) \propto \alpha_t(z_t)$.
- *Smoothing* $p(z_t \mid x_{0:T})$: belief about z_t given past *and* future readings (offline). Backward recursion: $\beta_t(z_t) = \sum_{z_{t+1}} p(x_{t+1} \mid z_{t+1})\, p(z_{t+1} \mid z_t)\, \beta_{t+1}(z_{t+1})$ with $\beta_T = 1$, and $p(z_t \mid x_{0:T}) \propto \alpha_t(z_t)\, \beta_t(z_t)$. At $t = T$ smoothing equals filtering.
- *Most probable path* $\arg\max_{z_{0:T}} p(z_{0:T} \mid x_{0:T})$ (Viterbi): same recursion as forward but with max instead of sum, plus backtracking.
- *In practice:* each recursion is evaluated for every value of z_t, so there is an implicit loop over all 10 states inside the loop over time, and the sum (or max) runs over the 10 previous states. The results are 10×100 matrices (states $\times$ time): $\alpha[i,t] = p(x_{0:t}, z_t = i)$ and $\beta[i,t] = p(x_{t+1:T} \mid z_t = i)$. In Viterbi, $\omega[i,t] = \max_{z_{0:t-1}} \log p(z_{0:t-1}, z_t = i, x_{0:t})$ is the log-probability of the best path ending in state i at time t, and $\psi[i,t]$ is the previous state on that path (a back-pointer). The path is read off backwards, starting from $\arg\max_i \omega[i,T]$.

Question 4: Accuracy

Implementation: Took the argmax state per time step for the filtered and smoothed distributions and compared with the true simulated states; accuracy is the fraction of the 100 steps that match. The Viterbi path is compared directly.

```r
# --- Accuracy = fraction of time steps where estimate == true state ---
# Point estimate per step: argmax over states of each column
# accuracy filtered
filteredStates <- apply(filtered, 2, which.max)
filteredAcc <- mean(filteredStates == simulation$states)
print(filteredAcc)
```

```
## [1] 0.57
```

```r
#accuracy smoothed
smoothedStates <- apply(smoothed, 2, which.max)
smoothedAcc <- mean(smoothedStates == simulation$states)
print(smoothedAcc)
```

```
## [1] 0.72
```

```r
#accuracy probable path (Viterbi already gives a state sequence)
probablePathAcc <- mean(probablePath == simulation$states)
print(probablePathAcc)
```

```
## [1] 0.65
```

Question 5: Repeat with a different sample

Implementation: Repeated Questions 3 and 4 on a new 100-step simulation (set.seed(222222)) from the same HMM and compared the three accuracies.

```r
# --- Same as Q3 + Q4 on a new simulated sample (new seed) ---
set.seed(222222)
# simHMM: list with $states (true path) and $observation (readings)
simulation <- simHMM(hmm, 100)
observations <- simulation$observation

filtered <- rep(0, 100)
smoothed <- rep(0, 100)

# Log alpha/beta, dims states x time
logAlpha <- forward(hmm, observations)
logBeta <- backward(hmm, observations)

# filtered: p(z_t | x_0:t) propto alpha_t (stable exp + normalize per column)
filtered <- apply(logAlpha, 2, function(col){
  col <- exp(col - max(col))
  col/sum(col)
})

#smoothed: p(z_t | x_0:T) propto alpha_t * beta_t
smoothed <- apply(logAlpha + logBeta, 2, function(col){
  col <- exp(col - max(col))
  col/sum(col)
})

#probable path (Viterbi, most probable joint sequence)
probablePath <- viterbi(hmm, observations)

#prints
print(filtered)
```

```
##       index
## states   1         2         3         4         5 6   7         8   9  10
```

```
##     1  0.0 0.0000000 0.0000000 0.00000000 0.0000000 1 0.5 0.0000000 0.0 0.0
##     2  0.0 0.0000000 0.0000000 0.00000000 0.0000000 0 0.5 0.6666667 0.4 0.2
##     3  0.0 0.0000000 0.0000000 0.00000000 0.0000000 0 0.0 0.3333333 0.6 0.5
##     4  0.2 0.1111111 0.0000000 0.00000000 0.0000000 0 0.0 0.0000000 0.0 0.3
##     5  0.2 0.2222222 0.0000000 0.00000000 0.0000000 0 0.0 0.0000000 0.0 0.0
##     6  0.2 0.2222222 0.2857143 0.00000000 0.0000000 0 0.0 0.0000000 0.0 0.0
##     7  0.2 0.2222222 0.2857143 0.33333333 0.1739130 0 0.0 0.0000000 0.0 0.0
##     8  0.2 0.2222222 0.2857143 0.33333333 0.3478261 0 0.0 0.0000000 0.0 0.0
##     9  0.0 0.0000000 0.1428571 0.25000000 0.3043478 0 0.0 0.0000000 0.0 0.0
##     10 0.0 0.0000000 0.0000000 0.08333333 0.1739130 0 0.0 0.0000000 0.0 0.0
##       index
## states   11        12        13         14         15        16        17
##     1  0.00 0.0000000 0.0000000 0.00000000 0.00000000 0.0000000 0.0000000
##     2  0.10 0.0000000 0.0000000 0.00000000 0.00000000 0.0000000 0.0000000
##     3  0.35 0.0000000 0.0000000 0.00000000 0.00000000 0.0000000 0.0000000
##     4  0.40 0.5172414 0.0000000 0.00000000 0.00000000 0.0000000 0.0000000
##     5  0.15 0.3793103 0.0000000 0.00000000 0.00000000 0.0000000 0.0000000
##     6  0.00 0.1034483 0.8235294 0.41176471 0.20588235 0.1272727 0.0000000
##     7  0.00 0.0000000 0.1764706 0.50000000 0.45588235 0.4090909 0.2864078
##     8  0.00 0.0000000 0.0000000 0.08823529 0.29411765 0.4636364 0.4660194
##     9  0.00 0.0000000 0.0000000 0.00000000 0.04411765 0.0000000 0.2475728
##     10 0.00 0.0000000 0.0000000 0.00000000 0.00000000 0.0000000 0.0000000
##       index
## states        18         19         20         21         22         23
##     1  0.0000000 0.06189320 0.15591735 0.23891547 0.29314067 0.32325244
##     2  0.0000000 0.00000000 0.00000000 0.07942584 0.16575320 0.25018109
##     3  0.0000000 0.00000000 0.00000000 0.00000000 0.00000000 0.00000000
##     4  0.0000000 0.00000000 0.00000000 0.00000000 0.00000000 0.00000000
##     5  0.0000000 0.00000000 0.00000000 0.00000000 0.00000000 0.00000000
##     6  0.0000000 0.00000000 0.00000000 0.00000000 0.00000000 0.00000000
##     7  0.1432039 0.07160194 0.03694427 0.00000000 0.00000000 0.00000000
##     8  0.3762136 0.25970874 0.17094552 0.10590112 0.05514034 0.03006157
##     9  0.3567961 0.36650485 0.32310582 0.25167464 0.18618170 0.13156465
##     10 0.1237864 0.24029126 0.31308704 0.32408293 0.29978409 0.26494024
##       index
## states         24         25        26        27        28        29        30
##     1  0.29858430 0.57771993 0.4751136 0.3483813 0.8055814 0.8372274 0.5000000
##     2  0.29109211 0.00000000 0.3032373 0.3891755 0.0000000 0.0000000 0.4186137
##     3  0.12699945 0.00000000 0.0000000 0.1516187 0.0000000 0.0000000 0.0000000
##     4  0.00000000 0.00000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000
##     5  0.00000000 0.00000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000
##     6  0.00000000 0.00000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000
##     7  0.00000000 0.00000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000
##     8  0.00000000 0.00000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000
##     9  0.08204633 0.09482575 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000
##     10 0.20127781 0.32745431 0.2216491 0.1108245 0.1944186 0.1627726 0.0813863
##       index
## states        31        32         33         34        35        36        37
##     1  0.3030242 0.1515121 0.07575604 0.03894004 0.0000000 0.0000000 0.0000000
##     2  0.4787903 0.3909072 0.27120966 0.17834694 0.0000000 0.0000000 0.0000000
##     3  0.2181855 0.3484879 0.36969758 0.32943846 0.0000000 0.0000000 0.0000000
##     4  0.0000000 0.1090928 0.22879034 0.30763412 0.5154361 0.2738525 0.1369263
##     5  0.0000000 0.0000000 0.05454638 0.14564044 0.3667307 0.4686974 0.3712750
##     6  0.0000000 0.0000000 0.00000000 0.00000000 0.1178333 0.2574501 0.3630737
```

```
##     7  0.0000000 0.0000000 0.00000000 0.00000000 0.0000000 0.0000000 0.1287250
##     8  0.0000000 0.0000000 0.00000000 0.00000000 0.0000000 0.0000000 0.0000000
##     9  0.0000000 0.0000000 0.00000000 0.00000000 0.0000000 0.0000000 0.0000000
##     10 0.0000000 0.0000000 0.00000000 0.00000000 0.0000000 0.0000000 0.0000000
##       index
## states        38       39        40        41 42  43   44    45     46
##     1  0.0000000 0.000000 0.0000000 0.0000000  0 0.0 0.00 0.125 0.2500
##     2  0.0000000 0.000000 0.0000000 0.0000000  0 0.0 0.00 0.000 0.0625
##     3  0.0000000 0.000000 0.0000000 0.0000000  0 0.0 0.00 0.000 0.0000
##     4  0.0992596 0.000000 0.0000000 0.0000000  0 0.0 0.00 0.000 0.0000
##     5  0.3684016 0.000000 0.0000000 0.0000000  0 0.0 0.00 0.000 0.0000
##     6  0.5323388 0.628535 0.0000000 0.0000000  0 0.0 0.00 0.000 0.0000
##     7  0.0000000 0.371465 0.7291473 0.0000000  0 0.0 0.00 0.000 0.0000
##     8  0.0000000 0.000000 0.2708527 0.7868732  1 0.5 0.25 0.125 0.0625
##     9  0.0000000 0.000000 0.0000000 0.2131268  0 0.5 0.50 0.375 0.2500
##     10 0.0000000 0.000000 0.0000000 0.0000000  0 0.0 0.25 0.375 0.3750
##       index
## states         47        48         49         50         51         52
##     1  0.38461538 0.0000000 0.00000000 0.21428571 0.33928571 0.36036036
##     2  0.00000000 0.0000000 0.00000000 0.00000000 0.10714286 0.22522523
##     3  0.00000000 0.0000000 0.00000000 0.00000000 0.00000000 0.05405405
##     4  0.00000000 0.0000000 0.00000000 0.00000000 0.00000000 0.00000000
##     5  0.00000000 0.0000000 0.00000000 0.00000000 0.00000000 0.00000000
##     6  0.00000000 0.0000000 0.00000000 0.00000000 0.00000000 0.00000000
##     7  0.00000000 0.0000000 0.00000000 0.00000000 0.00000000 0.00000000
##     8  0.03846154 0.1428571 0.07142857 0.03571429 0.01785714 0.00000000
##     9  0.19230769 0.8571429 0.50000000 0.28571429 0.16071429 0.09009009
##     10 0.38461538 0.0000000 0.42857143 0.46428571 0.37500000 0.27027027
##       index
## states         53 54  55   56        57         58        59         60
##     1  0.37837838  0 0.0 0.00 0.0000000 0.00000000 0.0000000 0.00000000
##     2  0.35135135  0 0.0 0.00 0.0000000 0.00000000 0.0000000 0.00000000
##     3  0.00000000  1 0.5 0.25 0.0000000 0.00000000 0.0000000 0.00000000
##     4  0.00000000  0 0.5 0.50 0.4285714 0.21428571 0.1363636 0.06818182
##     5  0.00000000  0 0.0 0.25 0.4285714 0.42857143 0.4090909 0.27272727
##     6  0.00000000  0 0.0 0.00 0.1428571 0.28571429 0.4545455 0.43181818
##     7  0.00000000  0 0.0 0.00 0.0000000 0.07142857 0.0000000 0.22727273
##     8  0.00000000  0 0.0 0.00 0.0000000 0.00000000 0.0000000 0.00000000
##     9  0.05405405  0 0.0 0.00 0.0000000 0.00000000 0.0000000 0.00000000
##     10 0.21621622  0 0.0 0.00 0.0000000 0.00000000 0.0000000 0.00000000
##       index
## states         61         62         63         64         65         66
##     1  0.00000000 0.00000000 0.00000000 0.0000000 0.06344086 0.14301075
##     2  0.00000000 0.00000000 0.00000000 0.0000000 0.00000000 0.03172043
##     3  0.00000000 0.00000000 0.00000000 0.0000000 0.00000000 0.00000000
##     4  0.03409091 0.00000000 0.00000000 0.0000000 0.00000000 0.00000000
##     5  0.17045455 0.10404624 0.00000000 0.0000000 0.00000000 0.00000000
##     6  0.35227273 0.26589595 0.00000000 0.0000000 0.00000000 0.00000000
##     7  0.32954545 0.34682081 0.40151515 0.2046332 0.00000000 0.00000000
##     8  0.11363636 0.22543353 0.37500000 0.3957529 0.33440860 0.16720430
##     9  0.00000000 0.05780347 0.18560606 0.2857143 0.37956989 0.35698925
##     10 0.00000000 0.00000000 0.03787879 0.1138996 0.22258065 0.30107527
##       index
## states         67         68         69         70         71         72
```

```
##     1  0.24760192 0.4039409 0.41862069 0.31763547 0.22517643 0.1224406
##     2  0.00000000 0.1627586 0.28334975 0.35098522 0.35345434 0.3146328
##     3  0.00000000 0.0000000 0.08137931 0.18236453 0.28194578 0.3455014
##     4  0.00000000 0.0000000 0.00000000 0.04068966 0.11791360 0.2174252
##     5  0.00000000 0.0000000 0.00000000 0.00000000 0.02150986 0.0000000
##     6  0.00000000 0.0000000 0.00000000 0.00000000 0.00000000 0.0000000
##     7  0.00000000 0.0000000 0.00000000 0.00000000 0.00000000 0.0000000
##     8  0.09322542 0.0000000 0.00000000 0.00000000 0.00000000 0.0000000
##     9  0.29226619 0.0000000 0.00000000 0.00000000 0.00000000 0.0000000
##     10 0.36690647 0.4333005 0.21665025 0.10832512 0.00000000 0.0000000
##       index
## states        73         74        75        76        77        78       79
##     1  0.0612203 0.04078331 0.0000000 0.0000000 0.0000000 0.0000000 0.000000
##     2  0.2185367 0.18636657 0.1159391 0.0000000 0.0000000 0.0000000 0.000000
##     3  0.3300671 0.36546507 0.2816593 0.2110327 0.2702315 0.2127419 0.106371
##     4  0.2814633 0.40738505 0.3944689 0.3588675 0.7297685 0.7872581 0.500000
##     5  0.1087126 0.00000000 0.2079326 0.3197357 0.0000000 0.0000000 0.393629
##     6  0.0000000 0.00000000 0.0000000 0.1103641 0.0000000 0.0000000 0.000000
##     7  0.0000000 0.00000000 0.0000000 0.0000000 0.0000000 0.0000000 0.000000
##     8  0.0000000 0.00000000 0.0000000 0.0000000 0.0000000 0.0000000 0.000000
##     9  0.0000000 0.00000000 0.0000000 0.0000000 0.0000000 0.0000000 0.000000
##     10 0.0000000 0.00000000 0.0000000 0.0000000 0.0000000 0.0000000 0.000000
##       index
## states        80        81         82       83        84        85        86
##     1  0.0000000 0.0000000 0.00000000 0.000000 0.0000000 0.0000000 0.0000000
##     2  0.0000000 0.0000000 0.00000000 0.000000 0.0000000 0.0000000 0.0000000
##     3  0.0000000 0.0000000 0.00000000 0.000000 0.0000000 0.0000000 0.0000000
##     4  0.0000000 0.0000000 0.00000000 0.000000 0.0000000 0.0000000 0.0000000
##     5  0.6942113 0.3471056 0.00000000 0.000000 0.0000000 0.0000000 0.0000000
##     6  0.3057887 0.5000000 0.51249834 0.360915 0.1804575 0.0000000 0.0000000
##     7  0.0000000 0.1528944 0.39500066 0.639085 0.5000000 0.0000000 0.0000000
##     8  0.0000000 0.0000000 0.09250099 0.000000 0.3195425 0.7194744 0.3597372
##     9  0.0000000 0.0000000 0.00000000 0.000000 0.0000000 0.2805256 0.5000000
##     10 0.0000000 0.0000000 0.00000000 0.000000 0.0000000 0.0000000 0.1402628
##       index
## states        87        88         89        90        91        92        93
##     1  0.0000000 0.0000000 0.17625159 0.0000000 0.0000000 0.2287772 0.3643886
##     2  0.0000000 0.0000000 0.00000000 0.0000000 0.0000000 0.0000000 0.1143886
##     3  0.0000000 0.0000000 0.00000000 0.0000000 0.0000000 0.0000000 0.0000000
##     4  0.0000000 0.0000000 0.00000000 0.0000000 0.0000000 0.0000000 0.0000000
##     5  0.0000000 0.0000000 0.00000000 0.0000000 0.0000000 0.0000000 0.0000000
##     6  0.0000000 0.0000000 0.00000000 0.0000000 0.0000000 0.0000000 0.0000000
##     7  0.0000000 0.0000000 0.00000000 0.0000000 0.0000000 0.0000000 0.0000000
##     8  0.2949936 0.1474968 0.07374841 0.1564969 0.0000000 0.0000000 0.0000000
##     9  0.7050064 0.5000000 0.32374841 0.8435031 0.5424455 0.2712228 0.1356114
##     10 0.0000000 0.3525032 0.42625159 0.0000000 0.4575545 0.5000000 0.3856114
##       index
## states         94         95         96         97         98         99
##     1  0.37500000 0.32716158 0.25250852 0.18534486 0.00000000 0.00000000
##     2  0.23938862 0.31623781 0.32741321 0.30167868 0.28536342 0.14268171
##     3  0.05719431 0.15265702 0.23861129 0.29444927 0.34929135 0.31732739
##     4  0.00000000 0.00000000 0.07768413 0.16453874 0.26893646 0.30911391
##     5  0.00000000 0.00000000 0.00000000 0.00000000 0.09640876 0.18267261
##     6  0.00000000 0.00000000 0.00000000 0.00000000 0.00000000 0.04820438
```

```
##     7  0.00000000 0.00000000 0.00000000 0.00000000 0.00000000 0.00000000
##     8  0.00000000 0.00000000 0.00000000 0.00000000 0.00000000 0.00000000
##     9  0.06780569 0.03490091 0.00000000 0.00000000 0.00000000 0.00000000
##     10 0.26061138 0.16904267 0.10378285 0.05398845 0.00000000 0.00000000
##       index
## states        100
##     1  0.00000000
##     2  0.00000000
##     3  0.00000000
##     4  0.44831974
##     5  0.35195254
##     6  0.16522971
##     7  0.03449801
##     8  0.00000000
##     9  0.00000000
##     10 0.00000000
```

```
print(smoothed)
```

```
##       index
## states     1    2   3    4 5 6         7         8         9       10        11
##     1  0.000 0.00 0.0 0.00 0 1 0.3549236 0.0000000 0.0000000 0.000000 0.0000000
##     2  0.000 0.00 0.0 0.00 0 0 0.6450764 0.7098472 0.1295416 0.000000 0.0000000
##     3  0.000 0.00 0.0 0.00 0 0 0.0000000 0.2901528 0.8704584 0.323854 0.0000000
##     4  0.000 0.00 0.0 0.00 0 0 0.0000000 0.0000000 0.0000000 0.676146 0.5181664
##     5  0.000 0.00 0.0 0.00 0 0 0.0000000 0.0000000 0.0000000 0.000000 0.4818336
##     6  0.125 0.00 0.0 0.00 0 0 0.0000000 0.0000000 0.0000000 0.000000 0.0000000
##     7  0.500 0.25 0.0 0.00 0 0 0.0000000 0.0000000 0.0000000 0.000000 0.0000000
##     8  0.375 0.75 0.5 0.00 0 0 0.0000000 0.0000000 0.0000000 0.000000 0.0000000
##     9  0.000 0.00 0.5 0.75 0 0 0.0000000 0.0000000 0.0000000 0.000000 0.0000000
##     10 0.000 0.00 0.0 0.25 1 0 0.0000000 0.0000000 0.0000000 0.000000 0.0000000
##       index
## states        12         13         14        15        16         17
##     1  0.0000000 0.00000000 0.00000000 0.0000000 0.0000000 0.00000000
##     2  0.0000000 0.00000000 0.00000000 0.0000000 0.0000000 0.00000000
##     3  0.0000000 0.00000000 0.00000000 0.0000000 0.0000000 0.00000000
##     4  0.0000000 0.00000000 0.00000000 0.0000000 0.0000000 0.00000000
##     5  0.7124788 0.00000000 0.00000000 0.0000000 0.0000000 0.00000000
##     6  0.2875212 0.90679117 0.55722744 0.2930745 0.1143324 0.00000000
##     7  0.0000000 0.09320883 0.42447025 0.5849100 0.5745282 0.48182938
##     8  0.0000000 0.00000000 0.01830232 0.1220154 0.3111394 0.44166667
##     9  0.0000000 0.00000000 0.00000000 0.0000000 0.0000000 0.07650395
##     10 0.0000000 0.00000000 0.00000000 0.0000000 0.0000000 0.00000000
##       index
## states         18          19          20         21         22         23
##     1  0.00000000 0.001122856 0.005482182 0.01649058 0.03885964 0.07859995
##     2  0.00000000 0.000000000 0.000000000 0.00000000 0.00000000 0.00000000
##     3  0.00000000 0.000000000 0.000000000 0.00000000 0.00000000 0.00000000
##     4  0.00000000 0.000000000 0.000000000 0.00000000 0.00000000 0.00000000
##     5  0.00000000 0.000000000 0.000000000 0.00000000 0.00000000 0.00000000
##     6  0.00000000 0.000000000 0.000000000 0.00000000 0.00000000 0.00000000
##     7  0.28386317 0.145640928 0.054730814 0.00000000 0.00000000 0.00000000
##     8  0.52008070 0.501348488 0.420651883 0.30797678 0.15972776 0.05950536
##     9  0.18306426 0.305808047 0.413780686 0.48382149 0.50056371 0.43862395
```

```
##     10 0.01299187 0.046079682 0.105354435 0.19171115 0.30084889 0.42327074
##       index
## states        24        25        26        27        28         29         30
##     1  0.1430211 0.2394326 0.3751441 0.5501556 0.7251671 0.90017863 0.45927874
##     2  0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.00000000 0.51565788
##     3  0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.00000000 0.00000000
##     4  0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.00000000 0.00000000
##     5  0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.00000000 0.00000000
##     6  0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.00000000 0.00000000
##     7  0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.00000000 0.00000000
##     8  0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.00000000 0.00000000
##     9  0.3199309 0.1403155 0.0000000 0.0000000 0.0000000 0.00000000 0.00000000
##     10 0.5370480 0.6202519 0.6248559 0.4498444 0.2748329 0.09982137 0.02506337
##       index
## states        31         32         33        34         35        36
##     1  0.1790412 0.03704301 0.00000000 0.0000000 0.00000000 0.0000000
##     2  0.5609072 0.36636108 0.13261547 0.0000000 0.00000000 0.0000000
##     3  0.2600516 0.48990807 0.51219260 0.3133893 0.00000000 0.0000000
##     4  0.0000000 0.10668784 0.32629731 0.5365199 0.60603655 0.2525152
##     5  0.0000000 0.00000000 0.02889462 0.1500908 0.35932712 0.6050503
##     6  0.0000000 0.00000000 0.00000000 0.0000000 0.03463633 0.1424345
##     7  0.0000000 0.00000000 0.00000000 0.0000000 0.00000000 0.0000000
##     8  0.0000000 0.00000000 0.00000000 0.0000000 0.00000000 0.0000000
##     9  0.0000000 0.00000000 0.00000000 0.0000000 0.00000000 0.0000000
##     10 0.0000000 0.00000000 0.00000000 0.0000000 0.00000000 0.0000000
##       index
## states         37        38        39        40 41 42        43        44
##     1  0.00000000 0.0000000 0.0000000 0.0000000  0  0 0.0000000 0.0000000
##     2  0.00000000 0.0000000 0.0000000 0.0000000  0  0 0.0000000 0.0000000
##     3  0.00000000 0.0000000 0.0000000 0.0000000  0  0 0.0000000 0.0000000
##     4  0.05050305 0.0000000 0.0000000 0.0000000  0  0 0.0000000 0.0000000
##     5  0.54775521 0.1874418 0.0000000 0.0000000  0  0 0.0000000 0.0000000
##     6  0.40174174 0.8125582 0.4582946 0.0000000  0  0 0.0000000 0.0000000
##     7  0.00000000 0.0000000 0.5417054 0.7291473  0  0 0.0000000 0.0000000
##     8  0.00000000 0.0000000 0.0000000 0.2708527  1  1 0.8461538 0.6923077
##     9  0.00000000 0.0000000 0.0000000 0.0000000  0  0 0.1538462 0.3076923
##     10 0.00000000 0.0000000 0.0000000 0.0000000  0  0 0.0000000 0.0000000
##       index
## states        45        46        47         48         49        50         51
##     1  0.0000000 0.0000000 0.0000000 0.00000000 0.00000000 0.2769231 0.58461538
##     2  0.0000000 0.0000000 0.0000000 0.00000000 0.00000000 0.0000000 0.09230769
##     3  0.0000000 0.0000000 0.0000000 0.00000000 0.00000000 0.0000000 0.00000000
##     4  0.0000000 0.0000000 0.0000000 0.00000000 0.00000000 0.0000000 0.00000000
##     5  0.0000000 0.0000000 0.0000000 0.00000000 0.00000000 0.0000000 0.00000000
##     6  0.0000000 0.0000000 0.0000000 0.00000000 0.00000000 0.0000000 0.00000000
##     7  0.0000000 0.0000000 0.0000000 0.00000000 0.00000000 0.0000000 0.00000000
##     8  0.5384615 0.3846154 0.2307692 0.07692308 0.01538462 0.0000000 0.00000000
##     9  0.4615385 0.6153846 0.7692308 0.92307692 0.43076923 0.1230769 0.00000000
##     10 0.0000000 0.0000000 0.0000000 0.00000000 0.55384615 0.6000000 0.32307692
##       index
## states        52 53 54        55        56         57        58         59
##     1  0.6153846  0  0 0.0000000 0.0000000 0.00000000 0.0000000 0.00000000
##     2  0.3846154  1  0 0.0000000 0.0000000 0.00000000 0.0000000 0.00000000
##     3  0.0000000  0  1 0.4733343 0.1589001 0.00000000 0.0000000 0.00000000
```

```
##     4  0.0000000  0  0 0.5266657 0.6288685 0.47670031 0.1801897 0.05377096
##     5  0.0000000  0  0 0.0000000 0.2122314 0.46660242 0.5930212 0.37925627
##     6  0.0000000  0  0 0.0000000 0.0000000 0.05669728 0.2267891 0.56697277
##     7  0.0000000  0  0 0.0000000 0.0000000 0.00000000 0.0000000 0.00000000
##     8  0.0000000  0  0 0.0000000 0.0000000 0.00000000 0.0000000 0.00000000
##     9  0.0000000  0  0 0.0000000 0.0000000 0.00000000 0.0000000 0.00000000
##     10 0.0000000  0  0 0.0000000 0.0000000 0.00000000 0.0000000 0.00000000
##       index
## states          60        61         62         63         64         65
##     1  0.000000000 0.0000000 0.00000000 0.00000000 0.00000000 0.01487076
##     2  0.000000000 0.0000000 0.00000000 0.00000000 0.00000000 0.00000000
##     3  0.000000000 0.0000000 0.00000000 0.00000000 0.00000000 0.00000000
##     4  0.009926781 0.0000000 0.00000000 0.00000000 0.00000000 0.00000000
##     5  0.175376735 0.0496339 0.00000000 0.00000000 0.00000000 0.00000000
##     6  0.522972289 0.3504798 0.15221064 0.00000000 0.00000000 0.00000000
##     7  0.291724195 0.4703526 0.47981245 0.35074625 0.10157647 0.00000000
##     8  0.000000000 0.1295336 0.32066511 0.46410677 0.48188495 0.29802154
##     9  0.000000000 0.0000000 0.04731181 0.17317751 0.34591876 0.49151355
##     10 0.000000000 0.0000000 0.00000000 0.01196946 0.07061982 0.19559415
##       index
## states         66        67         68         69         70          71
##     1  0.06704444 0.2081906 0.37325219 0.53573120 0.53247794 0.432929943
##     2  0.00000000 0.0000000 0.05779732 0.16102773 0.30996416 0.441302438
##     3  0.00000000 0.0000000 0.00000000 0.01154944 0.04622476 0.118993982
##     4  0.00000000 0.0000000 0.00000000 0.00000000 0.00123565 0.006773637
##     5  0.00000000 0.0000000 0.00000000 0.00000000 0.00000000 0.000000000
##     6  0.00000000 0.0000000 0.00000000 0.00000000 0.00000000 0.000000000
##     7  0.00000000 0.0000000 0.00000000 0.00000000 0.00000000 0.000000000
##     8  0.08046548 0.0000000 0.00000000 0.00000000 0.00000000 0.000000000
##     9  0.46449268 0.2522632 0.00000000 0.00000000 0.00000000 0.000000000
##     10 0.38799739 0.5395462 0.56895049 0.29169162 0.11009748 0.000000000
##       index
## states         72         73         74        75        76        77
##     1  0.23388710 0.10400812 0.03035756 0.0000000 0.0000000 0.0000000
##     2  0.51147588 0.46362606 0.33655928 0.1690818 0.0000000 0.0000000
##     3  0.23166679 0.37289454 0.50387095 0.5857897 0.5798445 0.3327713
##     4  0.02297023 0.05947128 0.12921221 0.2451285 0.4201555 0.6672287
##     5  0.00000000 0.00000000 0.00000000 0.0000000 0.0000000 0.0000000
##     6  0.00000000 0.00000000 0.00000000 0.0000000 0.0000000 0.0000000
##     7  0.00000000 0.00000000 0.00000000 0.0000000 0.0000000 0.0000000
##     8  0.00000000 0.00000000 0.00000000 0.0000000 0.0000000 0.0000000
##     9  0.00000000 0.00000000 0.00000000 0.0000000 0.0000000 0.0000000
##     10 0.00000000 0.00000000 0.00000000 0.0000000 0.0000000 0.0000000
##       index
## states         78        79        80         81       82        83        84
##     1  0.00000000 0.0000000 0.0000000 0.00000000 0.000000 0.0000000 0.0000000
##     2  0.00000000 0.0000000 0.0000000 0.00000000 0.000000 0.0000000 0.0000000
##     3  0.08569814 0.0000000 0.0000000 0.00000000 0.000000 0.0000000 0.0000000
##     4  0.91430186 0.4028267 0.0000000 0.00000000 0.000000 0.0000000 0.0000000
##     5  0.00000000 0.5971733 0.7199553 0.26804809 0.000000 0.0000000 0.0000000
##     6  0.00000000 0.0000000 0.2800447 0.65096496 0.654167 0.2054618 0.0000000
##     7  0.00000000 0.0000000 0.0000000 0.08098694 0.345833 0.7945382 0.5692804
##     8  0.00000000 0.0000000 0.0000000 0.00000000 0.000000 0.0000000 0.4307196
##     9  0.00000000 0.0000000 0.0000000 0.00000000 0.000000 0.0000000 0.0000000
```

```
##     10 0.00000000 0.0000000 0.0000000 0.00000000 0.000000 0.0000000 0.0000000
##       index
## states         85        86       87        88       89         90        91
##     1  0.00000000 0.0000000 0.000000 0.0000000 0.000000 0.00000000 0.0000000
##     2  0.00000000 0.0000000 0.000000 0.0000000 0.000000 0.00000000 0.0000000
##     3  0.00000000 0.0000000 0.000000 0.0000000 0.000000 0.00000000 0.0000000
##     4  0.00000000 0.0000000 0.000000 0.0000000 0.000000 0.00000000 0.0000000
##     5  0.00000000 0.0000000 0.000000 0.0000000 0.000000 0.00000000 0.0000000
##     6  0.00000000 0.0000000 0.000000 0.0000000 0.000000 0.00000000 0.0000000
##     7  0.00000000 0.0000000 0.000000 0.0000000 0.000000 0.00000000 0.0000000
##     8  0.93309899 0.7615155 0.589932 0.4183485 0.246765 0.07518146 0.0000000
##     9  0.06690101 0.2384845 0.410068 0.5816515 0.753235 0.92481854 0.4804023
##     10 0.00000000 0.0000000 0.000000 0.0000000 0.000000 0.00000000 0.5195977
##       index
## states        92         93         94          95         96         97
##     1  0.2663347 0.47365646 0.44995370 0.290535304 0.13770418 0.03886032
##     2  0.0000000 0.11764476 0.33510958 0.456356013 0.39281672 0.25300579
##     3  0.0000000 0.00000000 0.03758097 0.194877301 0.39037621 0.43214984
##     4  0.0000000 0.00000000 0.00000000 0.000000000 0.06778341 0.27598405
##     5  0.0000000 0.00000000 0.00000000 0.000000000 0.00000000 0.00000000
##     6  0.0000000 0.00000000 0.00000000 0.000000000 0.00000000 0.00000000
##     7  0.0000000 0.00000000 0.00000000 0.000000000 0.00000000 0.00000000
##     8  0.0000000 0.00000000 0.00000000 0.000000000 0.00000000 0.00000000
##     9  0.1801509 0.05617608 0.01355974 0.001937106 0.00000000 0.00000000
##     10 0.5535145 0.35252270 0.16379601 0.056294276 0.01131949 0.00000000
##       index
## states        98         99        100
##     1  0.0000000 0.00000000 0.00000000
##     2  0.1021118 0.00000000 0.00000000
##     3  0.3749614 0.22709891 0.00000000
##     4  0.3849348 0.44244167 0.44831974
##     5  0.1379920 0.26146341 0.35195254
##     6  0.0000000 0.06899601 0.16522971
##     7  0.0000000 0.00000000 0.03449801
##     8  0.0000000 0.00000000 0.00000000
##     9  0.0000000 0.00000000 0.00000000
##     10 0.0000000 0.00000000 0.00000000
```

```
print(probablePath)
```

```
##   [1]  6  7  8  9 10  1  1  2  2  3  4  5  6  6  7  8  9 10  1  1  1  1  1  1  1
##  [26]  1  1  1  1  1  1  1  2  3  4  4  4  5  6  7  8  8  8  8  8  8  8  9 10  1
##  [51]  1  1  2  3  3  3  4  4  5  6  7  8  9 10  1  1  1  1  1  1  1  1  1  1  2
##  [76]  3  3  3  4  5  5  6  6  7  8  8  8  8  8  9 10  1  1  1  1  1  1  2  3  4
```

```
# accuracy filtered (argmax per column vs true states)
filteredStates <- apply(filtered, 2, which.max)
filteredAcc <- mean(filteredStates == simulation$states)
print(filteredAcc)
```

```
## [1] 0.42
```

```r
#accuracy smoothed
smoothedStates <- apply(smoothed, 2, which.max)  # argmax state per column
smoothedAcc <- mean(smoothedStates == simulation$states)
print(smoothedAcc)
```

```
## [1] 0.46
```

```r
#accuracy probable path
probablePathAcc <- mean(probablePath == simulation$states)
print(probablePathAcc)
```

```
## [1] 0.39
```

ANSWER: The smoothing is in general better than filtered since smoothing has access to the entire observation sequence as opposed to filtered. Most probable path will in general be outperformed by smoothing because smoothing is maximizing P(zt | x0:T) separately for every t. Viterbi does it as a whole, Viterbi needs to respect the internal consistency of the transition model and cannot jump around implausibly among the states.

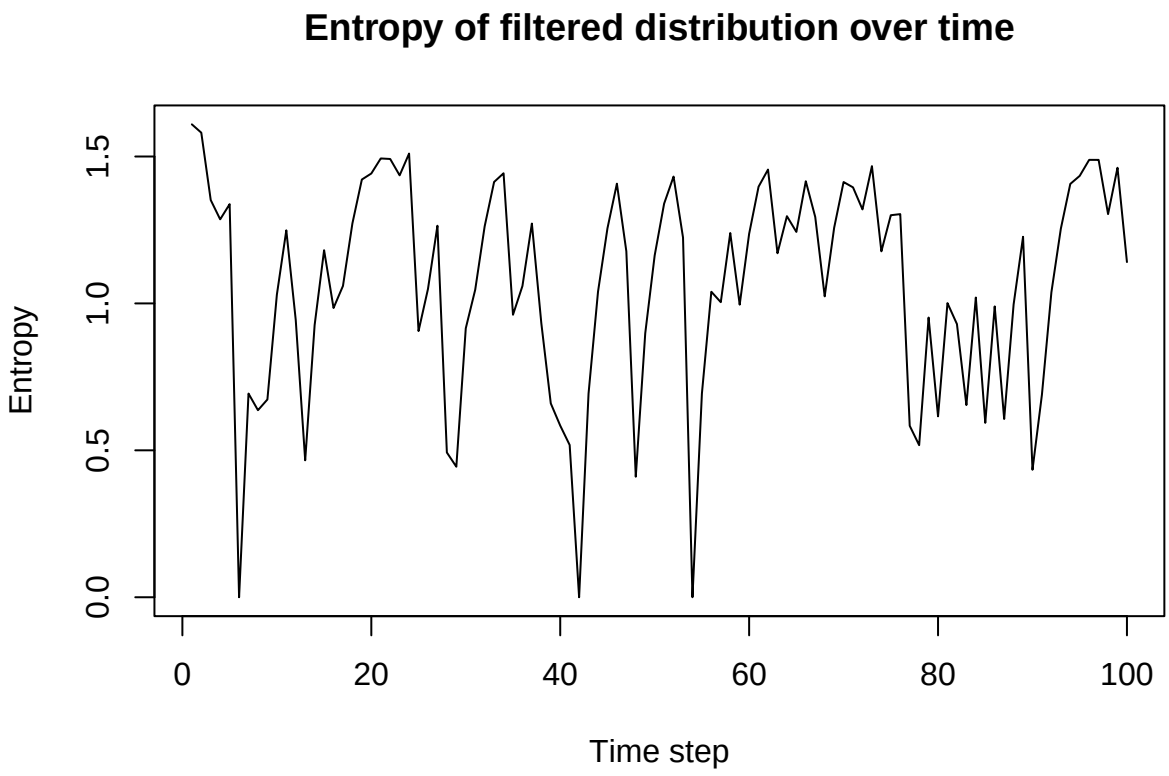
More precisely: (i) smoothing conditions on a superset of the data used by filtering, and conditioning on more data cannot increase the expected uncertainty about z_t; (ii) Taking the argmax of each smoothed marginal maximises the *expected number of correctly guessed states*, which is exactly what accuracy measures. Viterbi maximises the probability of the *whole* path being correct, a different objective, so it is not optimal per step. The price of smoothing is that its per-step argmaxes need not form a feasible path, e.g. it may guess sector 3 then sector 6, which the transition model forbids; Viterbi's path always has non-zero probability. Neither ordering is guaranteed on every single sample, only in expectation, which is why several samples should be compared.

Question 6: Entropy of the filtered distributions

Implementation: Computed entropy.empirical() (natural log) of the filtered distribution at each of the 100 time steps of the Question 5 sample and plotted it over time.

```r
# --- Entropy of filtered distribution at each time step ---
# entropy.empirical normalizes the column and computes -sum(p log p)
# (natural log). 0 = certain of state; log(10) = uniform/max uncertainty
filtered_entropy <- apply(filtered, 2, entropy.empirical)

plot(filtered_entropy, type = "l",
     xlab = "Time step", ylab = "Entropy",
     main = "Entropy of filtered distribution over time")
```



ANSWER: No. Each time step, the filtered distribution goes through two steps. The predict step pushes it through the transition model: the robot may stay or move, so the belief spreads out and the uncertainty grows. The update step weights it by the new reading, which shrinks the uncertainty, but by how much depends on the reading: some are more informative than others. When a step's prediction spreads the belief more than the new reading sharpens it, the entropy goes up. So uncertainty can increase at any time, even late in the sequence. Filtering still uses all past readings; the transition model just keeps adding new uncertainty about where the robot has gone.

Note that "more data cannot increase expected uncertainty" holds for a *fixed* variable, e.g. z_{50} given $x_{0:50}$ vs. $x_{0:100}$. Here the target z_t changes with t, so the filtered entropy fluctuates around a stationary level instead of decreasing towards 0; the plot shows this.

Question 7: Probabilities for time step 101

Implementation: Multiplied the filtered distribution at step 100 (Question 5 sample) by the transition matrix, giving $p(z_{101} \mid x_{0:100})$ as a vector over the 10 sectors.

```
# --- One-step prediction: p(z_101 | x_0:100) ---
# = sum_i p(z_100 = i | x_0:100) * p(z_101 | z_100 = i)
# i.e. row vector (filtered) %*% transProbs (rows = from-state)
filtered_t100 <- filtered[, 100]          # p(z_100 | x_0:100), length-10 vector

# %*% gives a 1x10 matrix; as.numeric drops dims -> plain length-10 vector
probs_101 <- as.numeric(filtered_t100 %*% transProbs)
names(probs_101) <- states
```

```r
print(probs_101)
```

```
##           1           2           3           4           5           6           7
## 0.00000000 0.00000000 0.00000000 0.22415987 0.40013614 0.25859113 0.09986386
##           8           9          10
## 0.01724900 0.00000000 0.00000000
```

Theory: This is one prediction step, i.e. filtering without the update (there is no x_{101}): $p(z_{101} \mid x_{0:100}) = \sum_{z_{100}} p(z_{101} \mid z_{100})\, p(z_{100} \mid x_{0:100})$. Each sector's mass is split half-half between staying and moving forward, so the prediction is never sharper than the filtered distribution. Predicting k steps ahead repeats the multiplication by the transition matrix and converges to the stationary (here uniform) distribution.